Rohan Kolla

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Education

Northeastern University, Boston, USA **May 2027**
Master of Science in Data Analytics Engineering **GPA: 4.0/4.0**
Relevant Courses: Data Management for Analytics, Foundations of Data Analytics Engineering, Computation and Visualization for Analytics, Data Mining in Engineering, Neural Networks and Deep Learning

Mahindra University, Hyderabad, India **Aug 2020 – Aug 2024**
Bachelor of Technology in Artificial Intelligence
Relevant Courses: Data Structures and Algorithms, Machine Learning, Reinforcement Learning, Optimization Techniques for AI, Natural Language Processing, Computational Sequence Modeling, Design and Analysis of Algorithms

Skills

Programming: Python, SQL, C/C++
AI/ML: PyTorch, TensorFlow, HuggingFace, Scikit-learn, NumPy, Pandas
LLM Systems: RAG, Prompt Engineering, LLM Evaluation, Structured Output Generation
Agent Systems: Multi-Agent Architectures, LangGraph, Tool-Augmented Reasoning
Data & Visualization: Tableau, Matplotlib, Plotly, Streamlit
Backend & APIs: FastAPI, Flask
Cloud & MLOps: AWS (EC2, S3), Docker, Kubernetes, CI/CD, Linux, Git

Work Experience

Random Trees, Hyderabad, India **Feb 2024 – Jun 2024**
AI/ML intern

- Built a generative AI pipeline using LLMs to extract structured information from industrial equipment documents, applying prompt engineering and reducing manual data processing time by **60%** enabling real-time analysis
- Developed reusable Python-based machine learning pipelines to automate model execution and deployment, reducing development time by **40%** while working closely with cross-functional stakeholders to translate business requirements into scalable AI solutions
- Built scalable AI infrastructure for edge and mobile development using TensorFlow Lite, Android Studio, and LLaVA models, improving computational efficiency while implementing model validation and AI safety mechanisms

Projects

HR Intelligence Platform **Jan 2026 – May 2026**

- Built an end-to-end multi-agent AI system (LangGraph + GPT-4o) to transform raw employee data into decision-ready insights, integrating semantic search, succession planning, and natural language analytics via FastAPI + Streamlit
- Designed a tool-augmented reasoning pipeline separating retrieval and response synthesis, achieving **95% retrieval accuracy**, reducing hallucinations, and improving latency by **85%** through parallel agent execution

Multimodal Image Captioning with CLIP and GPT-2 **Jan 2026 – May 2026**

- Architected a multimodal deep learning model combining **CLIP (vision encoder)** and **GPT-2 (language decoder)**, enabling high-quality image caption generation
- Applied **LoRA-based fine-tuning** on COCO dataset and conducted ablation studies, achieving **CIDEr score of 0.7668** and outperforming pretrained BLIP across evaluation metrics

Prompt Engineering Evaluation Platform **Nov 2025 – Jan 2026**

- Built an LLM evaluation platform using LangGraph multi-agent workflows to benchmark prompt strategies across models, enabling structured experimentation on reasoning, summarization, and Q&A tasks
- Developed reproducible pipelines with MLflow and Docker to analyze 500+ experiments, optimizing model outputs and reducing hallucination by **27%**

Real-Time Computer Vision Inference Platform **Oct 2025 – Dec 2025**

- Designed and deployed a production real-time computer vision system using YOLOv8 for people detection and counting, with FFmpeg and MediaMTX for low-latency HLS video streaming and a Flask-SocketIO backend powering a live React monitoring dashboard
- Architected a hybrid edge-cloud inference pipeline with local Mac Mini GPU inference and AWS EC2 orchestration, achieving sub-second end-to-end latency from camera capture to dashboard overlay across multiple concurrent streams